

8
SCREENS AUDITED

16
STRESS TESTS PASSED

5
LAYOUT OVERFLOWS FIXED

100%
PASS RATE (0 FAILURES)

Executive QA Summary

A full-scale automated test suite was executed across all user-facing views within the `mobile_user` application. Each screen was subjected to high-stress conditions, including **ultra-narrow viewports (300px width)**, **extreme font scaling (1.3x - 1.8x)**, **lengthy strings (50-100+ chars)**, **empty/null data fallbacks**, and **asynchronous platform lifecycle events**. All detected `RenderFlex` and pending timer issues were patched and verified.

Screen-by-Screen Test Coverage Matrix

Screen / Component	Edge Cases & Stress Conditions Tested	Discovered Behavior / Fix	Status
1. HomeView	300px narrow width, 1.5x font scale, 100-char protocol titles, empty continue-watching state.	Fixed 284px Header & 105px Resume Button collisions via <code>Expanded</code> & <code>FittedBox</code> .	PASSED
2. LibraryView	300px viewport, large font scaling, long category tags (PHASE 1 · PREPARATION), card navigation.	Fixed 246px Tag Pill & 140px Action Row overflow via <code>Flexible</code> + <code>TextOverflow.ellipsis</code> .	PASSED
3. CategoryDetailsView	Empty category video fallback, 80+ char video titles, in-progress & completed status indicators.	AppBar title wrapped in <code>Flexible</code> ; video cards handle long titles with graceful ellipsis.	PASSED
4. YoutubePlayerView	100+ character video titles, zero-second duration, 100% completed status, screen exit progress sync.	Resolved 3s memory leak Timer via cancellable lifecycle; guarded background Dio sync.	PASSED
5. NotificationSettingsView	Switch toggle state changes, dynamic time picker formatting (AM/PM), 300px narrow width.	Time cards and descriptive tip containers adapt fluidly	PASSED

Screen / Component	Edge Cases & Stress Conditions Tested	Discovered Behavior / Fix	Status
		across all mobile screen sizes.	
6. PersonalInfoView	50-char patient names, 80-char emails, 30-digit phone numbers, null profile fallback.	Fixed 993px column overflow using Expanded; null fields fallback to "Not provided".	PASSED
7. ProfileView	Extreme 99,999-day streaks, 14,500+ completed videos, 100+ platform hours, empty user state.	Stats row utilizes FittedBox and safe member-since date parsing without layout clipping.	PASSED
8. LoginView	300px narrow screen, password toggle visibility, loading indicator state, app update dialog.	Fixed 176px Password/Forgot-Password collision using Flexible & FittedBox.	PASSED

Discovered Layout Issues & Permanent Fixes

1. HomeView: Brand Header & Button RenderFlex Overflow

RESOLVED

Root Cause: Under narrow viewports (300px) or with 1.5x accessibility font scaling, the brand logo and title in HomeView overflowed by 284px because the inner Row was unconstrained. The "Resume video" button row also collided on high text scale.

```
- Row(children: [AppLogo(...), SizedBox(width: 10), Text('Meridian Health')]) + Expanded( +
  child: Row(children: [ + AppLogo(size: 38, iconSize: 20), + SizedBox(width: 10), +
    Flexible(child: Text('Meridian Health', overflow: TextOverflow.ellipsis, maxLines: 1)), + ], +
  )
```

2. LibraryView: Category Tag Pill & Action Row Overflow

RESOLVED

Root Cause: The colorful category cards in LibraryView contained fixed tag pills (e.g. PHASE 1 • PREPARATION) next to a 44px icon container without Flexible bounds, causing 246px horizontal clipping on 300px displays.

```
- Container(child: Text(tag, ...)) + Flexible( + child: Container( + child: Text(tag, overflow:
  TextOverflow.ellipsis, maxLines: 1, ...), + ), + )
```

3. LoginView: Password Label & Forgot Password Collision

RESOLVED

Root Cause: The header Row above the password field had both static "Password" text and a "Forgot Password?" TextButton without flex wrappers, causing a 176px overflow on narrow phone widths.

```
+ Flexible(child: Text('Password', overflow: TextOverflow.ellipsis)), + Flexible(child:
  Align(alignment: Alignment.centerRight, child: FittedBox(child: TextButton(...)))
```

4. YoutubePlayerView: 3-Second Timer Memory Leak & Api Sync Guard

RESOLVED

Root Cause: Future.delayed created an untracked 3s timer that lived past widget disposal. Additionally, ApiService().updateVideoProgress triggered unmocked background Dio requests on exit.

```
+ Timer? _fallbackTimer; + @override void dispose() { _fallbackTimer?.cancel();
  _subscription?.cancel(); super.dispose(); } + if (widget.enableApiSync) {
  ApiService().updateVideoProgress(...); }
```

Production Quality Recommendations

- **Accessibility:** All critical labels across Home, Library, and Profile now support font scaling up to 1.8x without crashing or visual clipping.
- **Network Fault Tolerance:** Network images and YouTube thumbnails are equipped with error fallback widgets.
- **State Resilience:** Profile and Personal Info screens provide clear "Loading..." and "Not provided" empty states when API responses are pending or empty.

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